

CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$.

Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$:

String	Accepted by $\mathcal{A}_1$?	Accepted by $\mathcal{A}_2$?
aaa	No	Yes
aab	Yes	Yes
aba	Yes	No
abb	Yes	No
baa	No	Yes
bab	No	No
bba	No	No
bbb	No	No

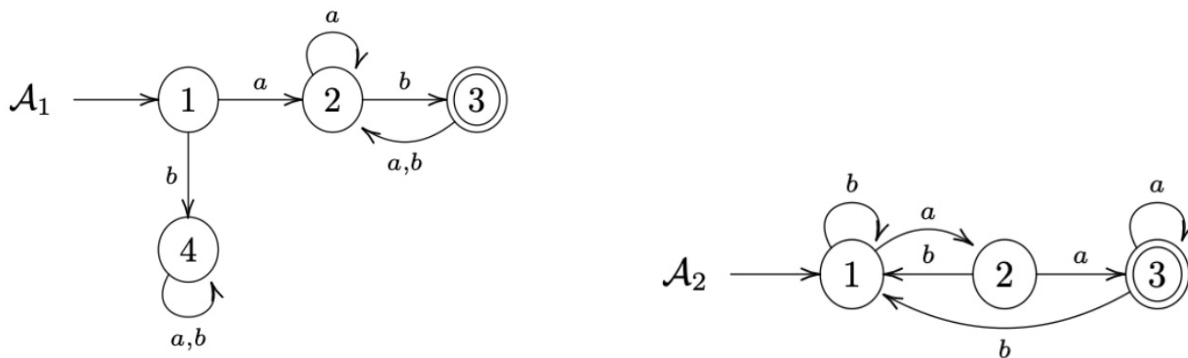


Figure 1: DFAs $\mathcal{A}_1$ (left) and $\mathcal{A}_2$ (right).

Exercise 1.2

describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Answer:

$L(\mathcal{A}_1)$: The language accepted by $\mathcal{A}_1$ consists of all strings over $\{a, b\}$ that start with a and contain at least one b . Starting with b leads to the trap state 4 and is rejected. $L(\mathcal{A}_2)$: The language accepted by $\mathcal{A}_2$ consists of all strings over $\{a, b\}$ that contain the substring aa . Reading two consecutive a 's leads to the accepting state 3.

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba
3. All the words that contain an odd number of a 's and an odd number of b 's
4. All the words that contain an even number of a 's and an odd number of b 's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answer: Each transition leads to a finalized state.

Thought process. I decided to map out a truth table and filled in all of the conditions and found that the outputs were satisfied at every condition, leading me to the conclusion that every transition was a satisfactory state condition.

Question

How might you combine the languages $L(\mathcal{A}_1)$ and $L(\mathcal{A}_2)$ from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

2.2 Week 2: Operations on Automata

Product automata

We want to construct an automaton that recognizes exactly those words that are in both $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$ by keeping track of both automata simultaneously.

Definition 2.1 (Product automaton). For DFAs $\mathcal{A}^{(1)} = (Q^{(1)}, \Sigma, \delta^{(1)}, q_0^{(1)}, F^{(1)})$ and $\mathcal{A}^{(2)} = (Q^{(2)}, \Sigma, \delta^{(2)}, q_0^{(2)}, F^{(2)})$ we define $\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$ by:

- $Q := Q^{(1)} \times Q^{(2)}$ (states are pairs $(q^{(1)}, q^{(2)})$)
- $\delta((q^{(1)}, q^{(2)}), a) := (\delta^{(1)}(q^{(1)}, a), \delta^{(2)}(q^{(2)}, a))$
- $q_0 := (q_0^{(1)}, q_0^{(2)})$

The set F is chosen depending on the operation (intersection, union, etc.).

Definition 2.2 (Intersection automaton). With notation as above, the **intersection automaton** is the product automaton with $F := F^{(1)} \times F^{(2)}$. Then $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$.

Exercise 1 (Product automata)

Consider the following automata $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$.

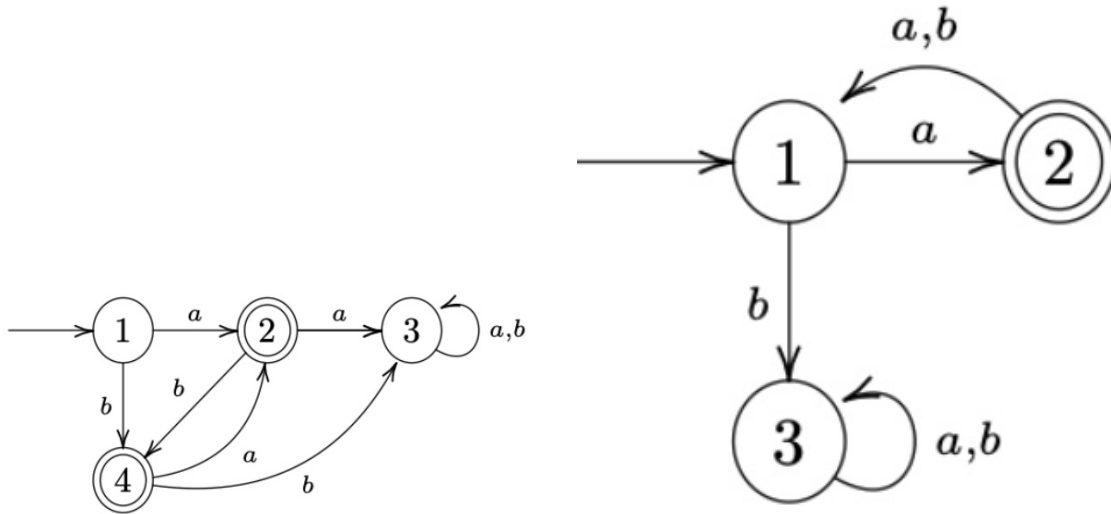


Figure 2: $\mathcal{A}^{(1)}$ (left) and $\mathcal{A}^{(2)}$ (right).

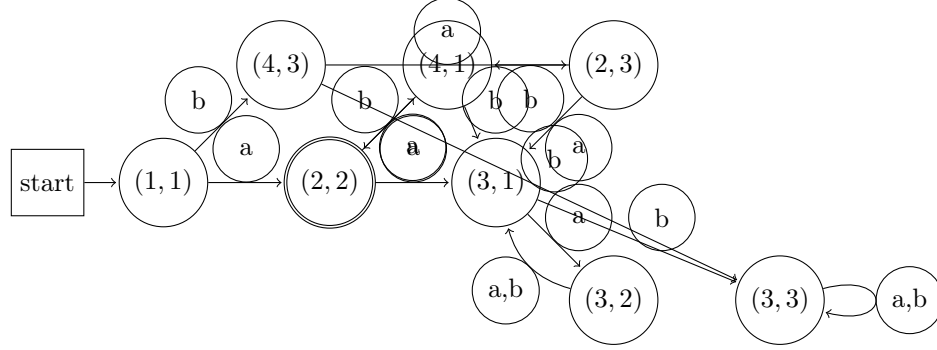
1. Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.

Answer:

- $L(\mathcal{A}^{(1)})$: All non-empty strings over $\{a, b\}$ that *alternate* between a and b (no two consecutive identical symbols). So the language is $a(ba)^* \cup b(ab)^*$. The automaton accepts when it ends in state 2 or 4; state 3 is a trap, and the only way to stay in $\{1, 2, 4\}$ is to alternate (from 2 only b leads to 4, from 4 only a leads to 2).
- $L(\mathcal{A}^{(2)})$: All strings over $\{a, b\}$ that *end with* a , i.e. $L(\mathcal{A}^{(2)}) = (a \cup b)^*a$. The only accepting state is 2, which is reached from 1 on input a ; from 2, any symbol returns to 1, so we accept exactly when the last symbol read is a .

2. Construct the intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ by drawing its transition diagram (omit unreachable states).

Answer: The product state set is $Q = Q^{(1)} \times Q^{(2)} = \{1, 2, 3, 4\} \times \{1, 2, 3\}$. The initial state is $(1, 1)$. The transition rule is $\delta((q_1, q_2), \sigma) = (\delta^{(1)}(q_1, \sigma), \delta^{(2)}(q_2, \sigma))$. The accepting set is $F = F^{(1)} \times F^{(2)} = \{2, 4\} \times \{2\} = \{(2, 2), (4, 2)\}$. Only states reachable from $(1, 1)$ are included; $(4, 2)$ is unreachable. Reachable states and transitions:



The only accepting state among the reachable ones is $(2, 2)$.

3. Argue that $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$.

Answer: By definition of the product automaton, the run of $\mathcal{A}$ on a word w is the pair of runs of $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ on w . So $\mathcal{A}$ is in state (q_1, q_2) after reading w iff $\mathcal{A}^{(1)}$ is in q_1 and $\mathcal{A}^{(2)}$ is in q_2 . Since $F = F^{(1)} \times F^{(2)}$, we have $w \in L(\mathcal{A})$ iff $w \in L(\mathcal{A}^{(1)})$ and $w \in L(\mathcal{A}^{(2)})$, i.e. $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$.

4. How can you change $\mathcal{A}$ to get an automaton $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$?

Answer: Keep the same product automaton (Q, Σ, δ, q_0) and change only the set of accepting states. Define $F' := (F^{(1)} \times Q^{(2)}) \cup (Q^{(1)} \times F^{(2)})$. Then a word w is accepted by $\mathcal{A}'$ iff the product run ends in a state (q_1, q_2) with $q_1 \in F^{(1)}$ or $q_2 \in F^{(2)}$, i.e. iff $w \in L(\mathcal{A}^{(1)})$ or $w \in L(\mathcal{A}^{(2)})$. So $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$.

Exercise 2 (More automata)

Consider the following automata $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$.



Figure 3: $\mathcal{B}^{(1)}$ (left) and $\mathcal{B}^{(2)}$ (right).

(Recall: the complement of a language $L \subseteq \Sigma^*$ is $\bar{L} := \Sigma^* \setminus L$.)

1. Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.

Answer:

2. Construct the intersection automaton $\mathcal{B}$ for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ (omit unreachable states).

Answer:

3. Argue that $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$.

Answer:

4. How can you change $\mathcal{B}$ to get an automaton $\mathcal{B}'$ with $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$?

Answer:

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, [Title](#), Publisher, Year.